In [ ]:

plot a line graph or a single graph for each property on the same graph

In [29]:
```
best_A = ((property_A['Daily_Revenue'].mean() + property_A['Daily_Revenue'].std()) * (property_A['Occupancy_Days'].mean() + pr
```

In [30]:
```
best_A.round(1)
```

Out[30]:
```
np.float64(7836992.4)
```

In [31]:
```
best_B = ((property_B['Daily_Revenue'].mean() + property_B['Daily_Revenue'].std()) * (property_B['Occupancy_Days'].mean() + pr
```

In [32]:
```
best_B.round(1)
```

Out[32]:
```
np.float64(7305016.7)
```

In [33]:
```
best_C = ((property_C['Daily_Revenue'].mean() + property_C['Daily_Revenue'].std()) * (property_C['Occupancy_Days'].mean() + pr
```

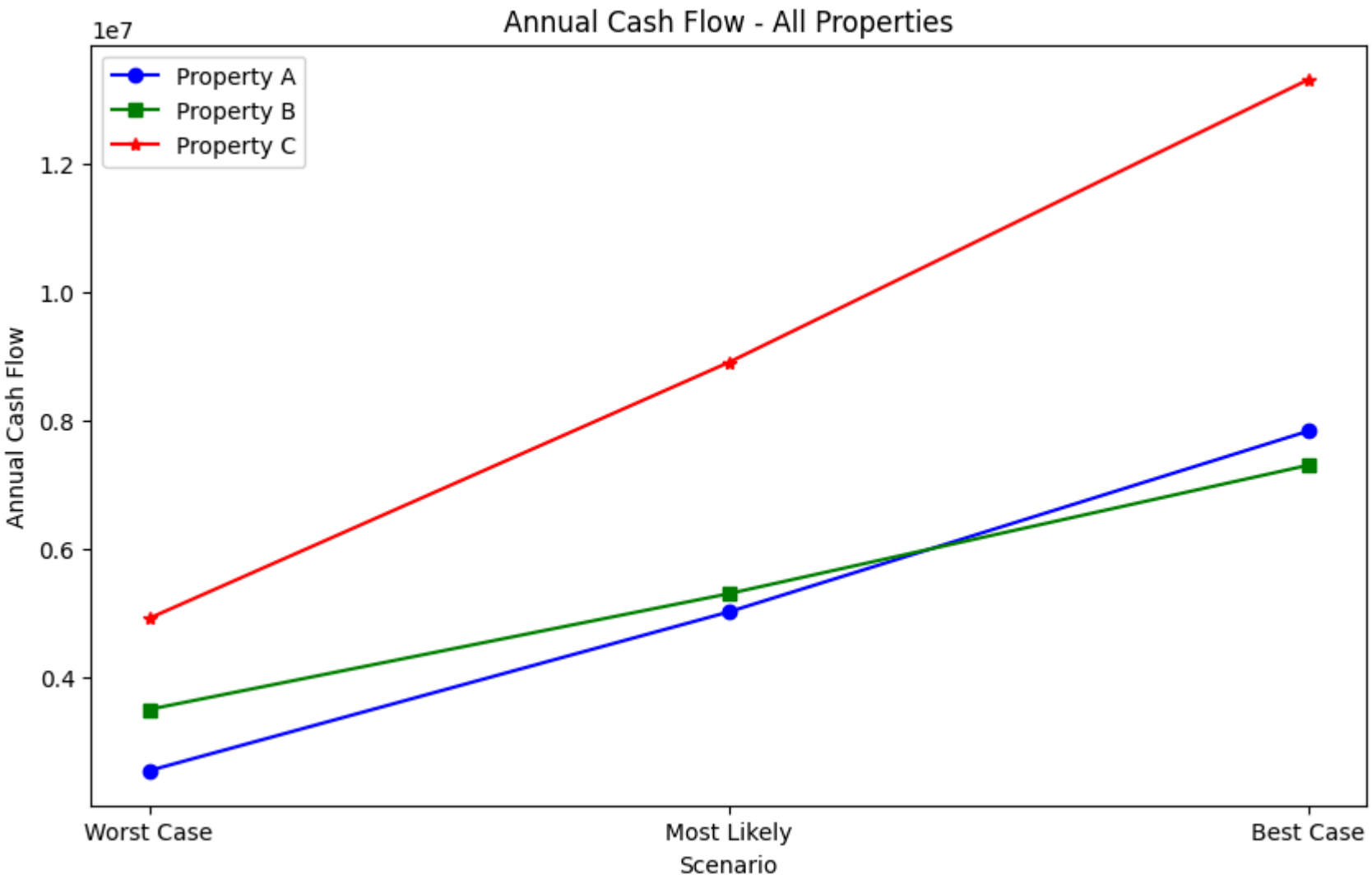
In [34]:
```
best_C.round(1)
```

Out[34]:
```
np.float64(13304468.0)
```

In [35]:
```
import matplotlib.pyplot as plt
```

In [36]:
```
worst_A = 2552275.2
revenue_A = 5022354.8
best_A = 7836992.4
worst_B = 3504875.9
revenue_B = 5305782.1
best_B = 7305016.7
worst_C = 4921432.6
revenue_C = 8903966.4
best_C = 13304468.0
scenarios = ['Worst Case', 'Most Likely', 'Best Case']
plt.figure(figsize=(10, 6))
plt.plot(scenarios, [worst_A, revenue_A, best_A], marker='o', label='Property A', color='blue')
plt.plot(scenarios, [worst_B, revenue_B, best_B], marker='s', label='Property B', color='green')
```

```
plt.plot(scenarios, [worst_C, revenue_C, best_C], marker='*', label='Property C', color='red')
plt.xlabel('Scenario')
plt.ylabel('Annual Cash Flow')
plt.title('Annual Cash Flow - All Properties')
plt.legend()
plt.show()
```



In [37]:
```
likely_daily_A = property_A['Daily_Revenue'].mean().round(1)
```

In [38]:
```
likely_daily_A
```

Out[38]:
```
np.float64(33507.7)
```

In [39]:
```
worst_daily_A = property_A['Daily_Revenue'].mean() - property_A['Daily_Revenue'].std()
```

In [40]:
```
worst_daily_A.round(1)
```

Out[40]:
```
np.float64(28701.6)
```

In [41]:
```
best_daily_A = property_A['Daily_Revenue'].mean() + property_A['Daily_Revenue'].std()
```

In [42]:
```
best_daily_A.round(1)
```

Out[42]:
```
np.float64(38313.8)
```

In [43]:
```
likely_daily_B = property_B['Daily_Revenue'].mean().round(1)
```

In [44]:
```
likely_daily_B
```

Out[44]:
```
np.float64(26090.9)
```

In [45]:
```
worst_daily_B = property_B['Daily_Revenue'].mean() - property_B['Daily_Revenue'].std()
```

In [46]:
```
worst_daily_B.round(1)
```

Out[46]:
```
np.float64(23153.3)
```

In [47]:
```
best_daily_B = property_B['Daily_Revenue'].mean() + property_B['Daily_Revenue'].std()
```

In [48]:
```
best_daily_B.round(1)
```

Out[48]:
```
np.float64(29028.4)
```

In [49]:
```
likely_daily_C = property_C['Daily_Revenue'].mean().round(1)
```

In [50]:
```
likely_daily_C
```

Out[50]:
```
np.float64(78450.9)
```

In [51]:
```
worst_daily_C = property_C['Daily_Revenue'].mean() - property_C['Daily_Revenue'].std()
```

In [52]:
```
worst_daily_C.round(1)
```

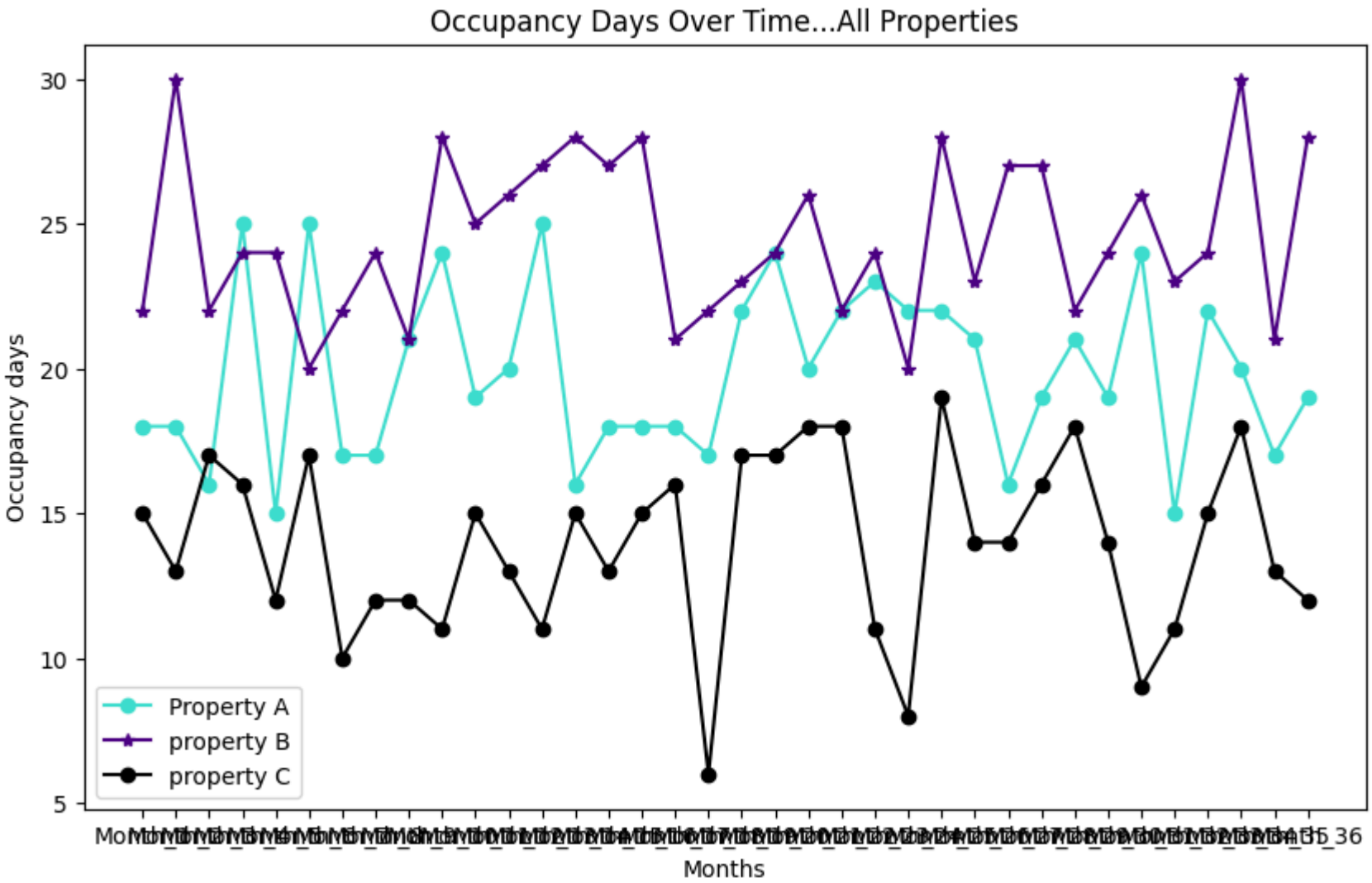
Out[52]:
```
np.float64(72853.0)
```

In [53]:
```
best_daily_C = property_C['Daily_Revenue'].mean() + property_C['Daily_Revenue'].std()
```

In [54]:
```
best_daily_C.round(1)
```

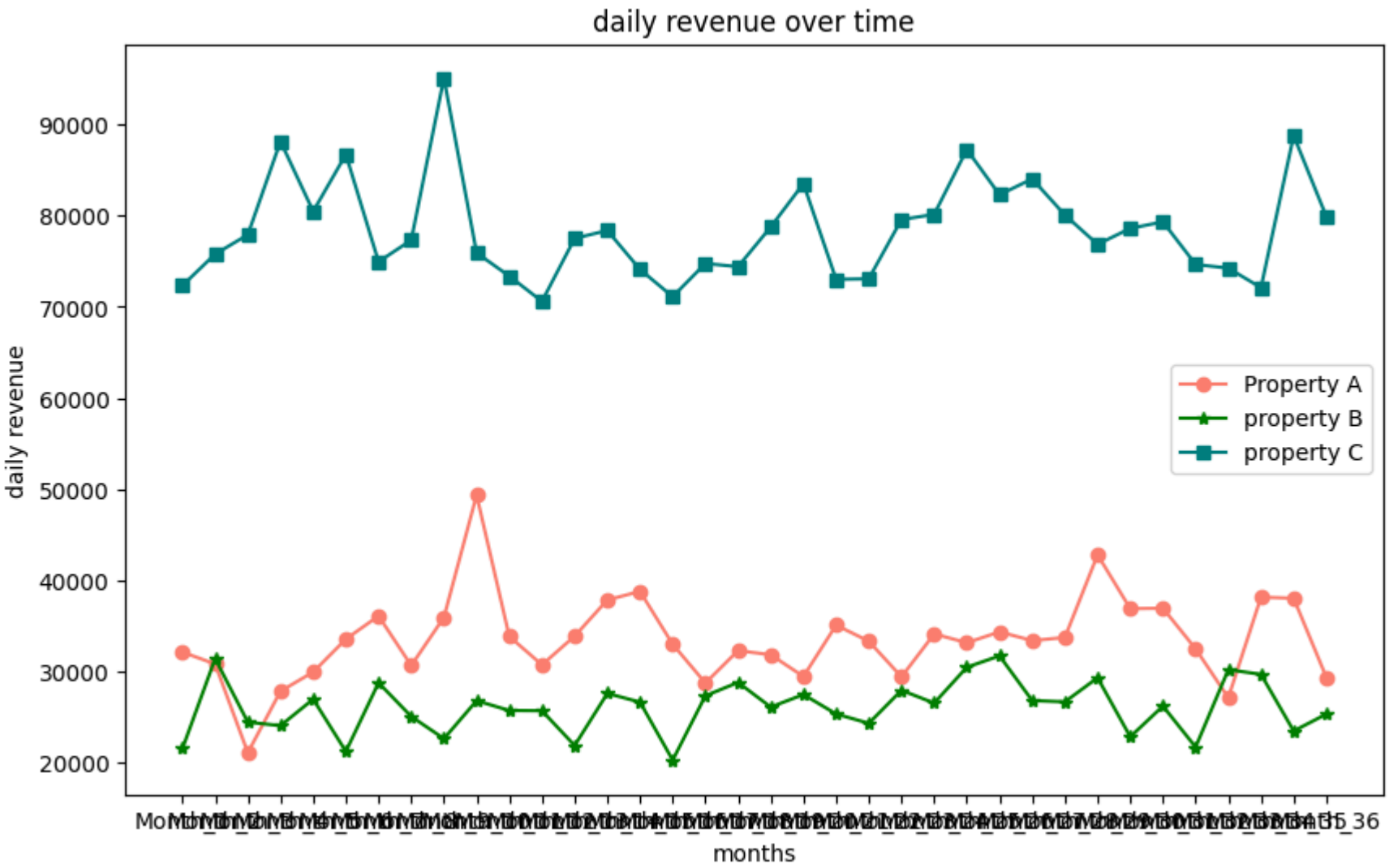
Out[54]:
```
np.float64(84048.8)
```

In [55]:
```
plt.figure(figsize=(10, 6))
plt.plot(property_A['Month'], property_A['Occupancy_Days'], marker='o', label='Property A', color='turquoise')
plt.plot(property_B['Month'], property_B['Occupancy_Days'], marker='*', label='property B', color='indigo')
plt.plot(property_C['Month'], property_C['Occupancy_Days'], marker='o', label='property C', color= 'k')
plt.xlabel('Months')
plt.ylabel('Occupancy days')
plt.title('Occupancy Days Over Time...All Properties')
plt.legend()
plt.show()
```



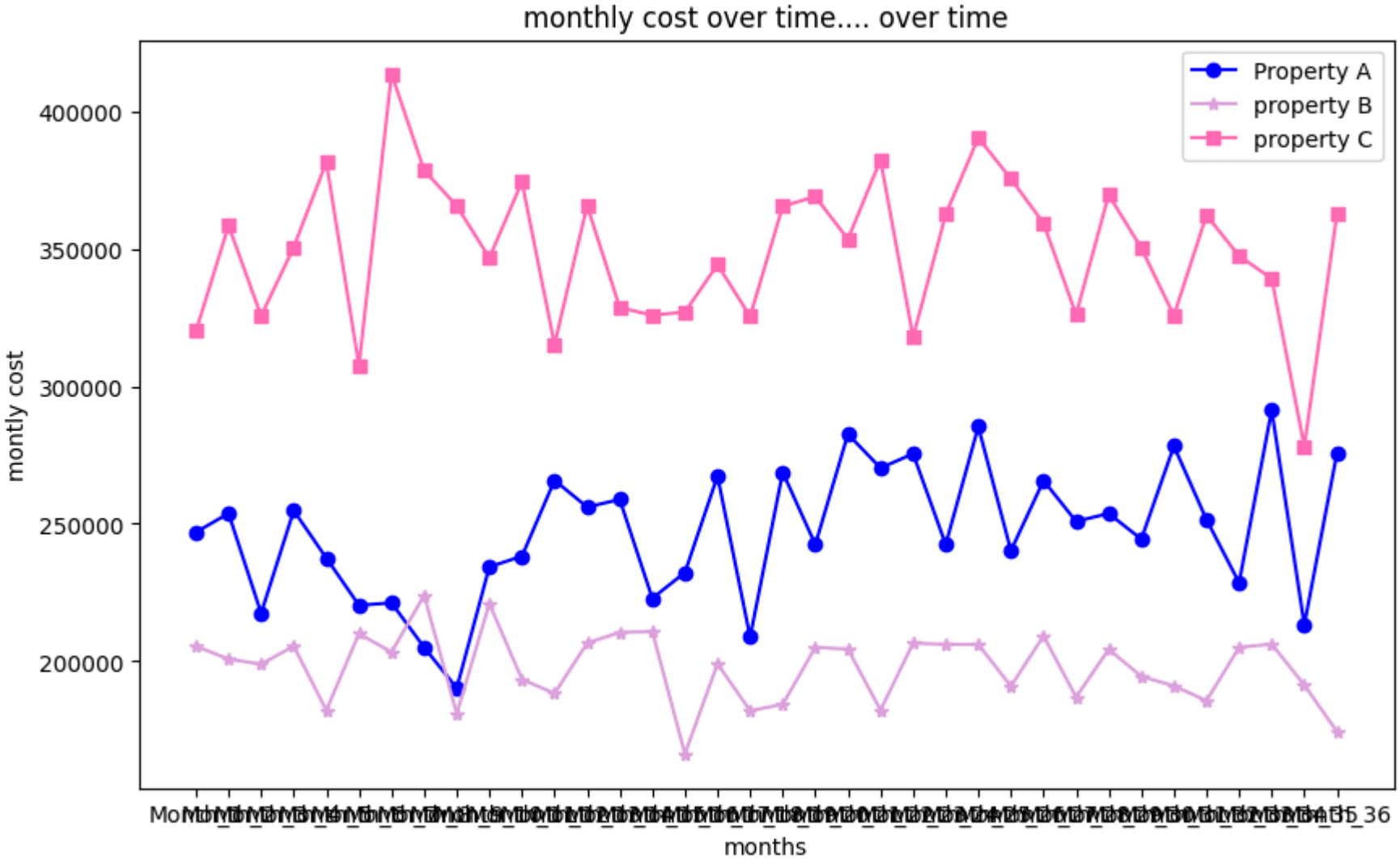
In [56]:
```
plt.figure(figsize=(10, 6))
plt.plot(property_A['Month'], property_A['Daily_Revenue'], marker='o', label='Property A', color='salmon')
plt.plot(property_B['Month'], property_B['Daily_Revenue'], marker='*', label='property B', color='green')
plt.plot(property_C['Month'], property_C['Daily_Revenue'], marker='s', label='property C', color= 'teal')
plt.xlabel('months')
plt.ylabel('daily revenue')
plt.title('daily revenue over time')
```

```
plt.legend()
plt.show()
```



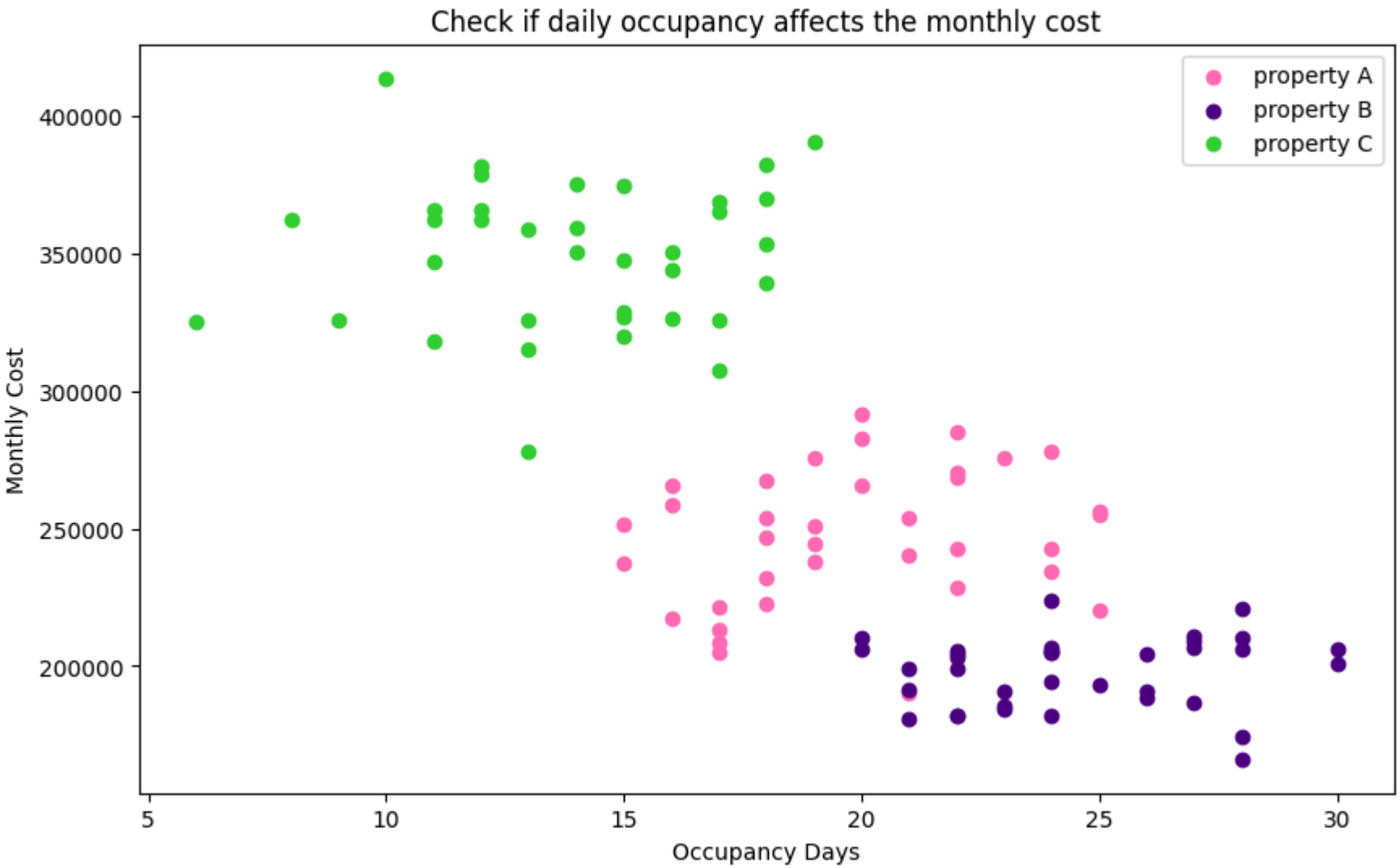
In [57]:
```
plt.figure(figsize=(10, 6))
plt.plot(property_A['Month'], property_A['Monthly_Cost'], marker='o', label='Property A', color='blue')
plt.plot(property_B['Month'], property_B['Monthly_Cost'], marker='*', label='property B', color='plum')
plt.plot(property_C['Month'], property_C['Monthly_Cost'], marker='s', label='property C', color= 'hotpink')
plt.xlabel('months')
```

```
plt.ylabel('montly cost')
plt.title('monthly cost over time.... over time')
plt.legend()
plt.show()
```



In [65]:
```
plt.figure(figsize=(10, 6))
plt.scatter(property_A['Occupancy_Days'], property_A['Monthly_Cost'], color='hotpink', label='property A')
plt.scatter(property_B['Occupancy_Days'], property_B['Monthly_Cost'], color='indigo', label='property B')
```

```
plt.scatter(property_C['Occupancy_Days'], property_C['Monthly_Cost'], color='limegreen', label='property C')
plt.xlabel('Occupancy Days')
plt.ylabel('Monthly Cost')
plt.title('Check if daily occupancy affects the monthly cost')
plt.legend()
plt.show()
```



In [80]:
```
aff_a= property_A['Occupancy_Days'].corr(property_A['Monthly_Cost'])
aff_b= property_B['Occupancy_Days'].corr(property_B['Monthly_Cost'])
aff_c= property_C['Occupancy_Days'].corr(property_C['Monthly_Cost'])
```

In [82]:
```
aff_a
```

Out[82]:
```
np.float64(0.23257725507930613)
```

In [84]:
```
aff_b
```

Out[84]:
```
np.float64(0.08297711814003562)
```

In [85]:
```
aff_c
```

Out[85]:
```
np.float64(0.043145760757868926)
```

In [ ]: